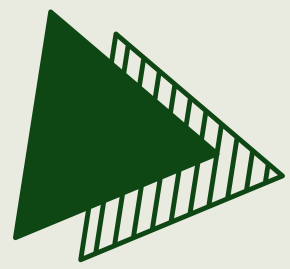


CREDGE AI

An End-to-End Multi Agent AI System for Personal Finance: Synthetic Data Generation, Budget Optimization, and Investment Advisory

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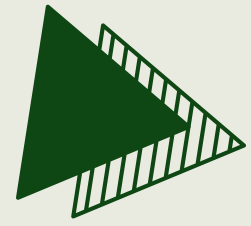
Credge AI

Problem

- Managing personal finances is complex and sensitive. Real transaction data are scarce due to privacy, making it hard to train models.
- At the same time, individuals need smarter tools to understand spending and investments, beyond basic rules or siloed apps.

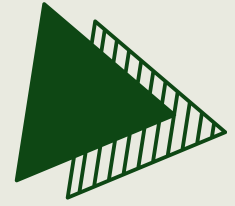
Solution Overview

- This project proposes an integrated AI-driven personal finance system spanning Synthetic Data Generation, Budget Classification & Optimization, and an LLM-Powered Investment Advisor.
- The goal is to bridge data gaps and provide a seamless, AI-enhanced budgeting and investment guidance platform.



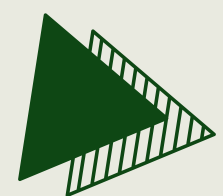
MOTIVATION & GOALS

- Privacy-Preserving Data: Need realistic synthetic financial data to develop models without exposing personal info.
- Smart Budgeting: Improve on generic guidelines by providing personalized, AI-driven spending adjustments.
- Informed Investing: Help users make investment decisions by analyzing holistic signals (market sentiment, fundamentals, etc.).
- Goal: Boost financial literacy and decision-making through an end-to-end system that guides from budgeting to investing.



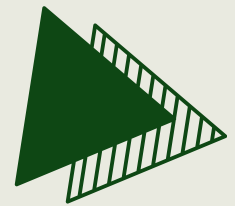
SYNTHETIC DATA GENERATOR

- Generates realistic transactions to mimic personal finance data (no real user data needed).
- Five-stage pipeline (statistical, rule-based, GAN, LLM) to progressively enhance data realism.
- Privacy & Utility: Synthetic dataset reflects real-world spending patterns while preserving privacy.
- Provides training data for downstream modules (classification & advisor) in absence of real data.

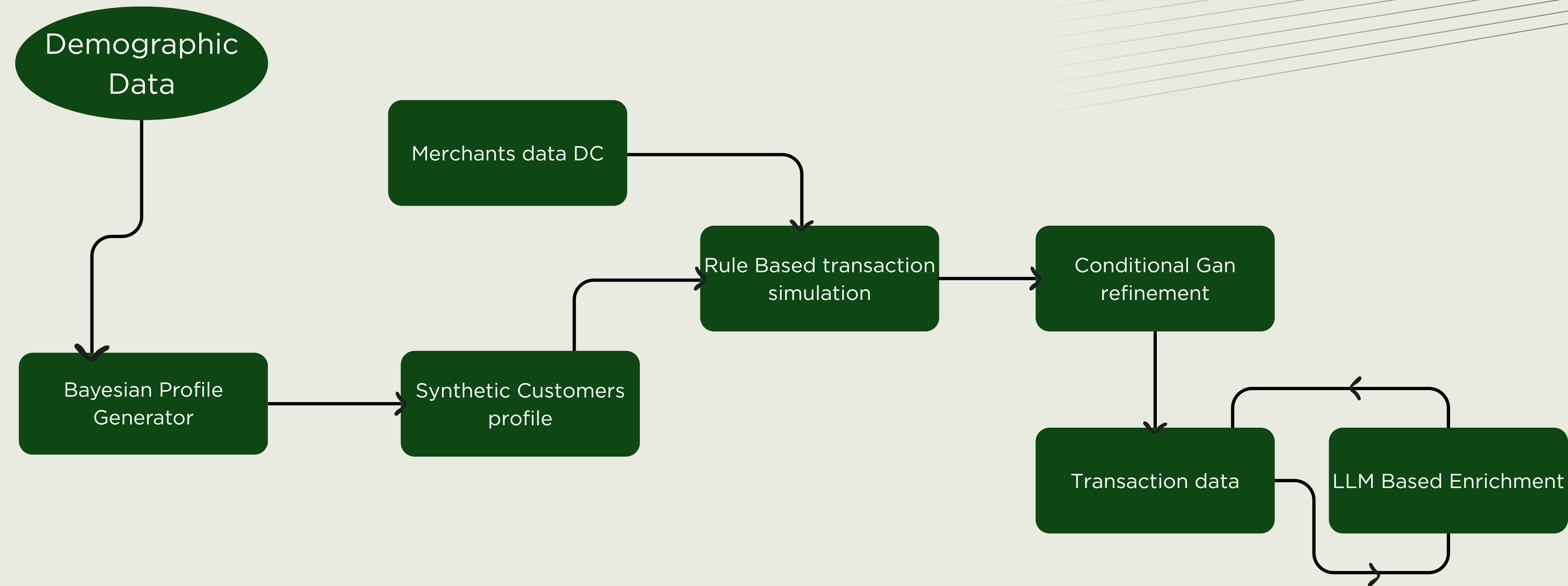


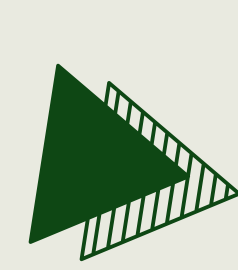
SYNTHETIC DATA GENERATOR – MULTI-STAGE PROCESS

- Stage 1: Bayesian Demographics – Sample synthetic households from census stats.
- Stage 2: Rule-Based Simulation – Simulate each household's transactions (allocate monthly budget to categories, generate purchases with plausible timing and merchants).
- Stage 3: GAN Refinement – Apply a Conditional Tabular GAN (CTGAN) to learn complex patterns from Stage 2 data (captures correlations like spending habits).
- Stage 4: LLM Enrichment – Use a language model to validate and enrich transactions (e.g. ensure merchant names and descriptions make sense, correct anomalies).
- Stage 5: Iterative Feedback Loop – Repeat GAN training with LLM-validated data for 2-3 cycles, so the generator progressively improves and produces high-plausibility data.



SYNTHETIC DATA GENERATOR – PIPELINE DIAGRAM

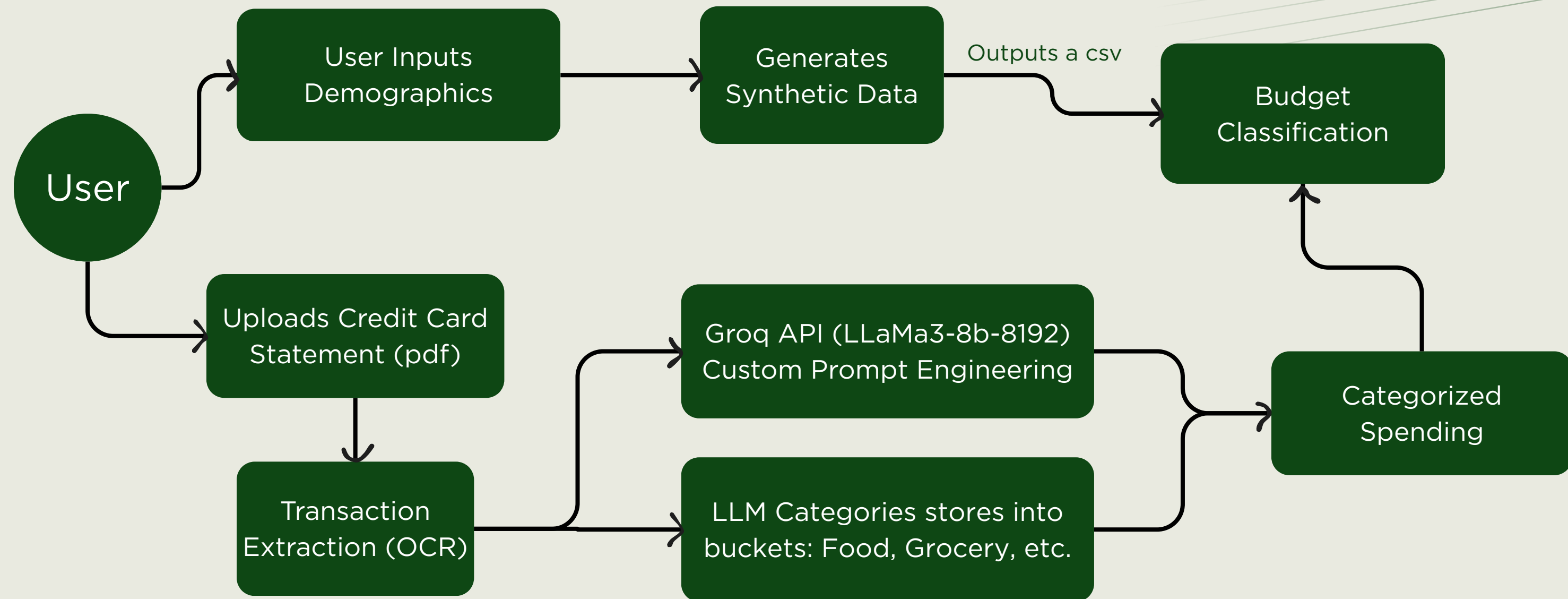


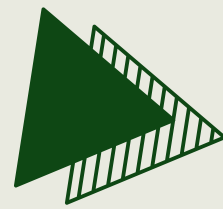


SYNTHETIC DATA GENERATOR — SAMPLE RESULTS

- Realistic output: The final synthetic dataset mirrors real spending distributions (categories, amounts, timing) as observed in actual financial data.
- Only ~2–3 feedback iterations were needed for the GAN to internalize the LLM’s domain knowledge, yielding highly plausible transactions.
- Examples: Generated customer profiles have sensible attributes (e.g. 30-year-old, \$75k income, family of 3) and their transactions look authentic (e.g. ~\$150 at a local grocery store).
- Privacy ensured: No real user data was used, yet models trained on this synthetic data perform well due to its realism.

BUDGET CLASSIFICATION





BUDGET CLASSIFICATION – MODELS

- **Automated Expense Categorization:**

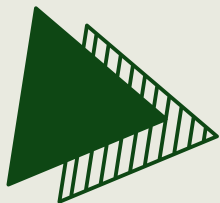
Classifies each transaction description into categories (e.g. Grocery, Rent, Travel, etc.).

- **Models explored:**

- *Logistic Regression (baseline bag-of-words).*
- *LSTM RNN (recurrent neural network).*
- *BERT-base-uncased (fine-tuned transformer for classification).*
- *LLaMA3-8b-8192 (large language model in few-shot mode).*

- **Dataset used for fine-tuning:**

- *Yelp Open Dataset*
- *Synthetic Data*



BUDGET CLASSIFICATION - ACCURACY COMPARISON

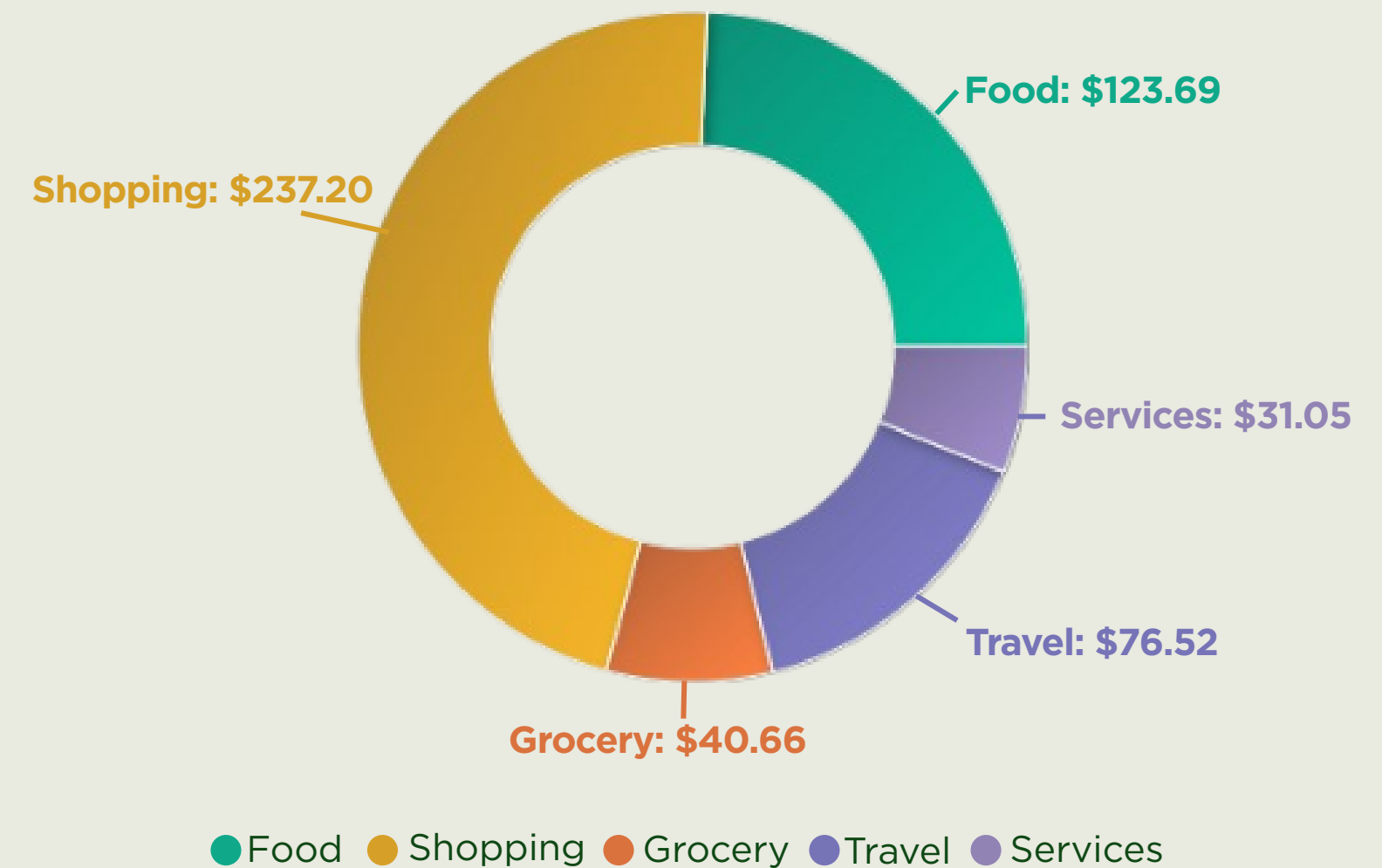
Model performance on the transaction categorization: *higher the better*

Model	Accuracy	Macro F1	Notes
Logistic Regression	71%	0.57	Struggled with edge cases (underrepresented categories)
LSTM	39%	0.40	Overfit the training data; poor generalization
Bert-base-uncase	72%	0.62	Balanced performance, improved recall for minority class
LLaMa3-8b-8192 (few- shot)	N/A	N/A	High accuracy observed qualitatively (human eval); no training needed

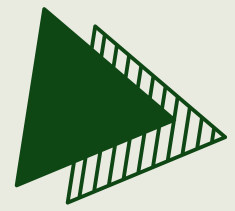
Performance: Fine-tuned BERT achieved the best balance (~72% accuracy, improved recall for rare classes), roughly on par with logistic regression’s accuracy but more robust. LSTM severely underperformed (~39% accuracy due to overfitting). The few-shot LLM showed high accuracy qualitatively without any training data.

► VISUALIZATION

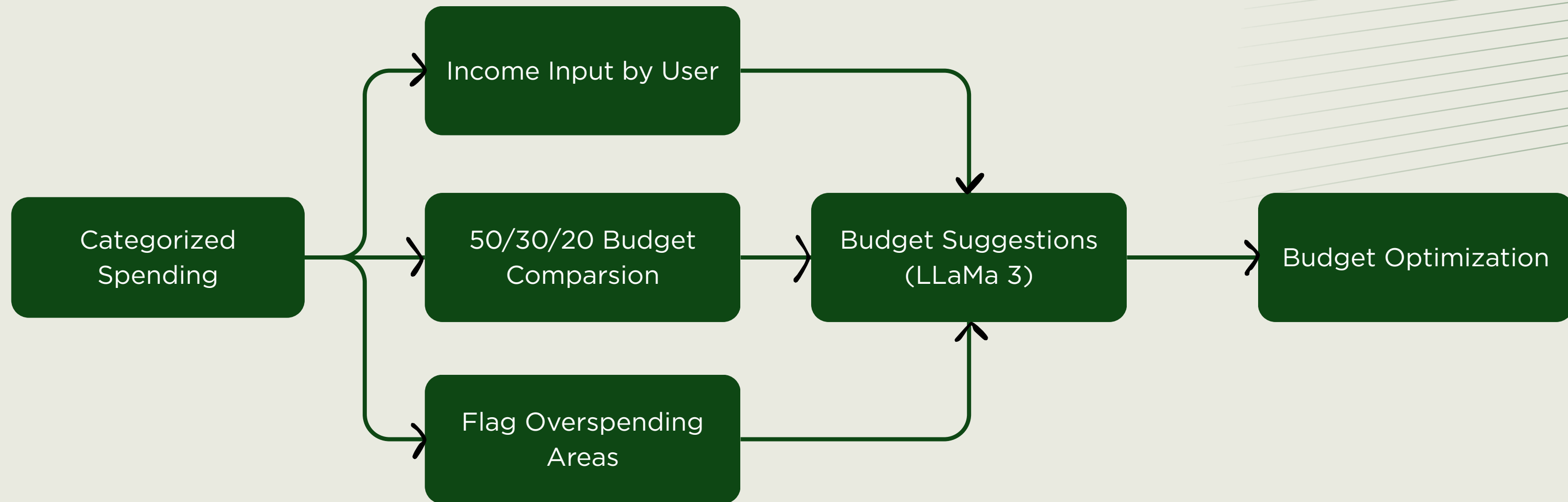
- The donut chart displays categorized transaction outputs from the Budget Classification Agent, segmented using LLM-based labeling.
- Provides structured inputs to the **Budget Optimization Agent** for downstream budget rule evaluation and adjustment inference.

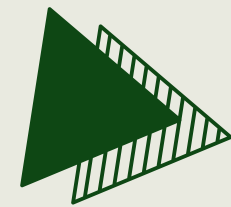


Categorized spending breakdown



BUDGET OPTIMIZATION

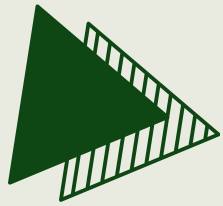




BUDGET OPTIMIZATION

- After **categorization**, the system evaluates spending against the **50/30/20 budget rule***:
 - **Needs**: 50% of income (essentials like rent, groceries, utilities).
 - **Wants**: 30% (discretionary: entertainment, dining, travel, etc.).
 - **Savings**: 20% (savings or debt payments).
- Each spending category is mapped to these buckets (e.g. *Rent* → *Needs*, *Travel* → *Wants*, etc.).
- The agent flags areas exceeding recommended limits (e.g. if “**Wants**” > **30%** of income, or “**Needs**” > **50%**).
- Uses an **LLM-powered output** to suggest adjustments. It generates a friendly message with tips to reduce spending in the over-budget categories.
- **Example Suggestions**: Switch to cheaper alternatives for groceries, cut unused subscriptions – typically identifying ~**3-5%** of income that could be saved through optimizations.

*Based on: E. Warren and A. Tyagi, *All Your Worth: The Ultimate Lifetime Money Plan*, Free Press, 2005.



BUDGET OPTIMIZATION

Bucket	Target (%)	Target (\$)
Needs	50	700
Wants	30	420
Savings	20	280

Sample: Monthly Income (\$1400)

✓ Food - \$77.98 (Under Control)

⚠ Travel - \$181.44 (Over Budget)
Travel is eating into your budget. Try off-season bookings or loyalty points.

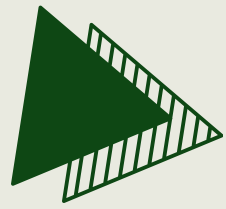
✓ Services - \$31.93 (Under Control)

✓ Shopping - \$105.93 (Under Control)

Flag Overspending Areas

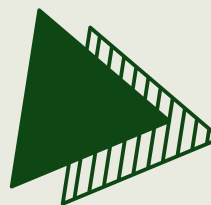
Category	Issue	Recommendation	Estimated Monthly Savings
Grocery	Spending above essential budget	Shop at discount supermarkets; plan meals and buy in bulks to reduce costs.	\$45 (approx)
Services	Multiple subscriptions (high total)	Cancel or downgrade unnecessary subscriptions; negotiate lower rates for utilities/internet.	\$25(approx)

Recommendations: Example Scenario



INVESTMENT ADVISOR: TECHNICAL INDICATORS

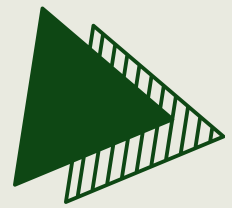
- Technical Analysis Module: Evaluates market momentum for stocks using standard indicators:
- **RSI (Relative Strength Index)**: signals if a stock is overbought (>70) or oversold (<30).
- **MACD (Moving Average Convergence Divergence)**: two moving averages' crossover indicates trend shifts.
- **Moving Averages (50-day & 200-day)**: crossovers (e.g. golden cross) show trend direction and momentum.
- **OBV (On-Balance Volume)**: measures buying/selling pressure via volume trends; ATR (Average True Range): gauges volatility.
- The advisor computes these from price data and combines them into a technical score (e.g. +1 for each bullish signal, -1 for bearish conditions).
- Influence: Technical signals adjust timing – e.g. if fundamentals are good but technicals are very bearish, the system might hold off on a Buy (wait for a better entry). If technicals confirm positive news, it strengthens the recommendation.



INVESTMENT ADVISOR: FUNDAMENTAL ANALYSIS

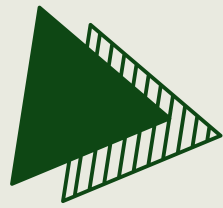
- **Fundamental Analysis Module:** Assesses a company's financial health and valuation:
 - Looks at metrics like revenue growth (↑ good), P/E ratio (is it reasonable vs peers?), debt levels (manageable or high?).
 - Determines if a company has **strong fundamentals** (e.g. consistent growth, undervalued stock, low debt) or **weak fundamentals** (declining sales, overvalued, high debt).
- Optionally parses qualitative reports (earnings calls, management discussion) via LLM for additional insight, but primarily focuses on **quantitative factors**.

Outcome: Produces a fundamental score or summary. **Strong fundamentals** bias the agent toward a **“Buy”** recommendation (assuming other factors aren't negative), whereas poor fundamentals might trigger caution or a **“Sell”** if issues are significant.



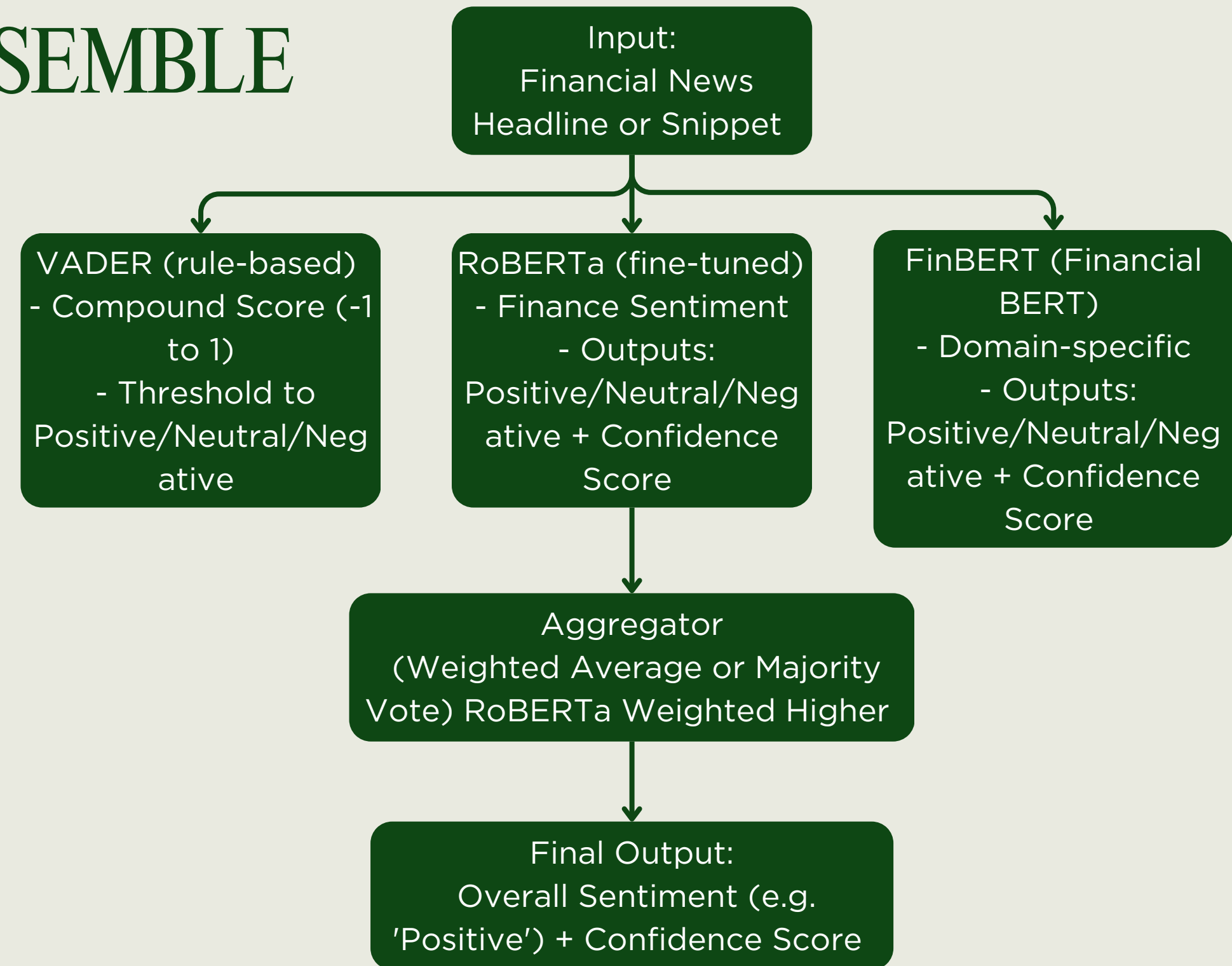
INVESTMENT ADVISOR: DECISION ENGINE & INTEGRATION

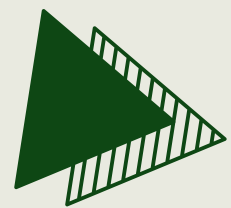
- **Multi-Signal Fusion:** A central decision engine combines **sentiment**, **fundamental**, and **technical** signals for each stock.
- **Risk Tolerance Aware:** Adjusts advice based on user's risk profile – e.g. a conservative investor gets safer recommendations than an aggressive investor, even on the same data.
- **Decision Logic:** Uses simple rules from our findings:
 - If most signals are strongly positive (and model predicts high chance of stock rise) $\Rightarrow$ output **“Buy (High Confidence)”**.
 - If signals are mixed or moderate $\Rightarrow$ output **“Hold/Neutral”**.
 - If multiple signals are negative or low upward probability $\Rightarrow$ output **“Sell”** or avoid recommending.
- For **portfolio requests**, the engine selects a mix of stocks aligned with the risk level (more stable assets for low risk, more growth for high risk) and assigns weights to each.
- Finally, a **GPT-based Explanation module** generates a user-friendly explanation of the recommendation, outlining the reasoning (e.g. “Stock X is recommended because... based on strong earnings and positive market sentiment”).



SENTIMENT ANALYSIS ENSEMBLE

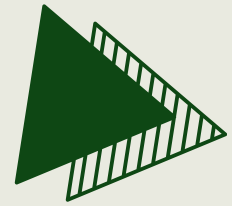
- **News Sentiment Module:** Uses an ensemble of llm/nlp models to gauge market sentiment from financial news:
- **FinBERT:** Finance-specific BERT model giving Positive/Neutral/Negative sentiment tailored to financial text.
- **RoBERTa:** A transformer model fine-tuned on finance news, captures nuanced language and context.
- **VADER:** A lexicon + rule-based sentiment tool (outputs a polarity score, which we map to Positive/Neutral/Negative).
- Each relevant news headline is fed through **all three models**. The results are then combined:
- FinBERT's output gets higher weight (expert on finance context).
- RoBERTa helps catch subtler cues or sarcasm that FinBERT might miss.
- VADER provides a simple sentiment check and breaks ties if the others disagree.
- The ensemble yields an overall sentiment label (e.g. "Overall sentiment: Positive") along with a confidence score.
- Benefit: This multi-model approach is more reliable than any single model, reducing errors. For example, if one model is off, the other two can override it – leading to a more robust sentiment signal for the advisor.



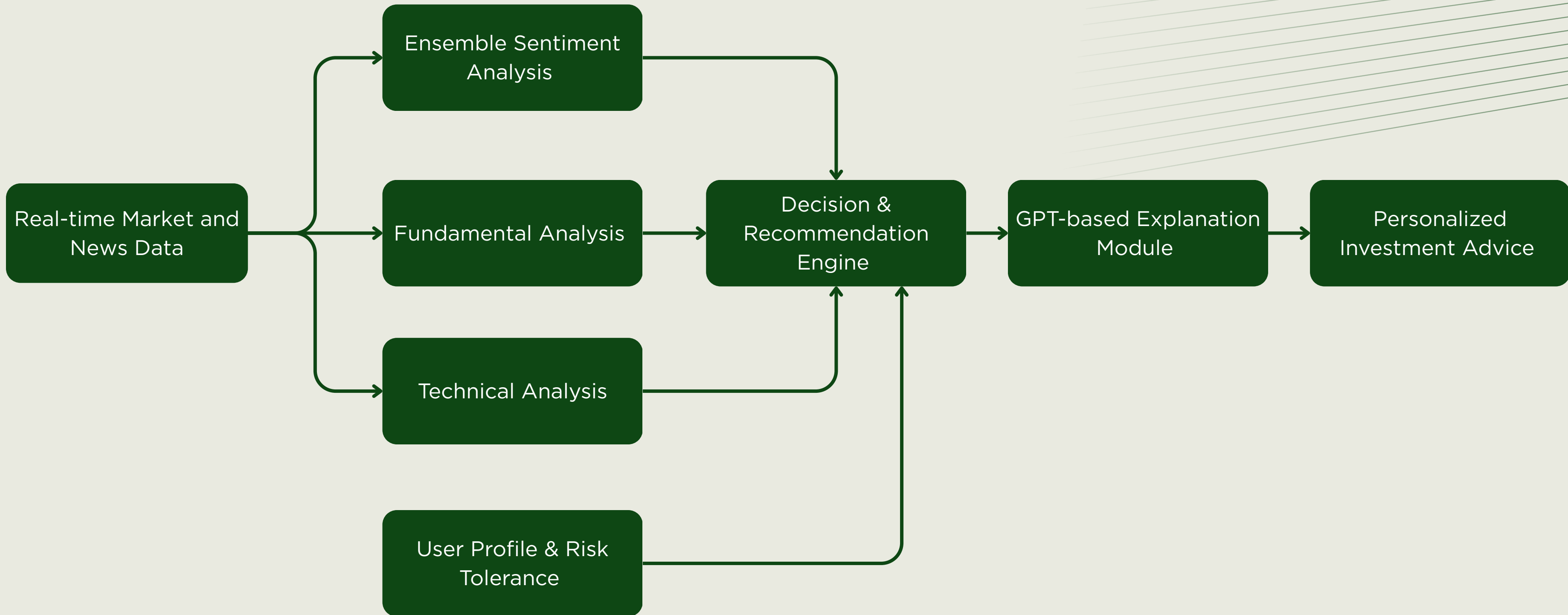


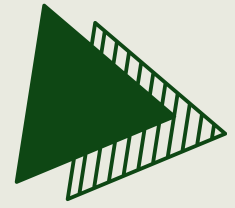
INVESTMENT RECOMMENDATIONS & PORTFOLIO OUTPUT

- **Single-Stock Recommendation:** e.g. “Buy” or “Hold” a specific stock, with a confidence level. The system weighs all signals and responds (e.g. “Buy Apple – strong fundamentals and positive momentum” or “Hold – mixed signals”).
- **Portfolio Recommendation:** Construct a sample portfolio based on a target risk level (Conservative, Moderate, Aggressive).
- **Risk-Tailored Portfolio Examples:**
 - *Low-Risk:* Emphasize stable stocks. *E.g.*, 40% in JNJ (Johnson & Johnson), 30% in KO (Coca-Cola), 20% in WMT (Walmart), 10% in MSFT (Microsoft) – these are blue-chip, dividend-paying companies for steady growth.
 - *High-Risk:* Emphasize growth and momentum. *E.g.*, 25% TSLA, 20% NVDA, 20% COIN, 20% GME, 15% AMD – higher volatility picks with potential for big gains (and bigger risk).
- The agent allocates weights (summing to 100%). Higher-confidence picks get larger weights, ensuring a balanced but optimized portfolio suggestion.
- **Explanation Provided:** For any recommendation, the **LLM explanation** module will justify the choices (e.g., “Chosen for its strong earnings and positive news sentiment” for each stock in the portfolio). This helps users trust and understand the advice.



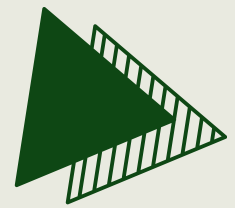
LLM-ADVISOR ARCHITECTURE





CONCLUSION

- We developed an integrated multi-agent AI system addressing key personal finance challenges:
- Created a synthetic data generator that produces lifelike financial data, enabling model training without privacy issues.
- Built a budget classification & optimization assistant that can automatically categorize spending and suggest ~**3-5%** income in potential savings via personalized tips.
- Implemented an LLM-driven investment advisor that synthesizes sentiment, fundamentals, and technical analysis into actionable stock recommendations and portfolio allocations (achieving notable accuracy, e.g. sentiment classification accuracy up to ~68% F1).
- Overall, the system supports users in budgeting better and investing smarter with AI-driven insights and explanations.



FUTURE WORK

- **Enhance Accuracy:** Fine-tune classification and advisory models (e.g., train the LLM on finance-specific data) to improve performance and reliability. Reduce need for manual review of AI suggestions.
- **User Testing:** Deploy the system to real users or larger simulations to gather feedback. Use this to refine the advice style and usefulness.
- **Expand Capabilities:** Include more financial instruments (like bonds, ETFs), integrate real-time data feeds for live advice, and handle more complex scenarios (e.g. tax implications, dynamic goals).
- **Automation & Safety:** Add automated checks on LLM outputs to prevent any incorrect or harmful financial advice. Ensure the recommendations remain sound and trustworthy.
- **Final Thought:** This project demonstrates the potential of combining multiple AI agents in personal finance. By uniting data generation, budgeting, and investment analysis, such systems can **empower individuals** to make informed financial decisions with confidence.



Thank you.